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Author(s): John S. Preisser, Andrzej T. Galecki, Kurt K. Lohman and Lynne E. Wagenknecht

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Analysis of Smoking Trends With Incomplete Longitudinal Binary Responses

John S. PREISSER, Andrzej T. GALECKI, Kurt K. LOHMAN, and Lynne E. WAGENKNECHT

The generalized estimating equations procedure (GEE) widely applied in the analysis of correlated binary data requires that missing data depend only on remote covariates or that they be missing completely at random (MCAR); otherwise GEE regression parameter estimates are biased. A weighted generalized estimating equations (WGEE) approach that accounts for dropouts under the less stringent assumption of missing at random (MAR) through dependence on observed responses gives unbiased estimation of parameters in the model for the marginal means if the dropout mechanism is specified correctly. WGEEs are applied in the estimation of 7-year trends in cigarette smoking in the United States from a cohort of 5,078 black and white young adults. Analysis using WGEE suggests that there was a general decline in cigarette smoking only among white females, whereas the only other subgroup for which smoking declined was white males of the older birth cohort (1955–1962) with college degrees. The results of WGEE are compared to a likelihood-based method valid under MAR that does not require specification of a missing data model.

KEY WORDS: Dropouts; Maximum likelihood; Missing at random; Weighted generalized estimating equations.

1. INTRODUCTION

Despite an intensive antismoking social and political climate in the United States in the 1980s and 1990s, reductions in smoking have not met national health objectives (U.S. Department of Health and Human Services 1991). This article presents an analysis of trends in cigarette smoking based on data from the Coronary Artery Risk Development in young Adults (CARDIA) study, an epidemiological cohort study that recorded cardiovascular risk factors on five occasions over a 10-year period in black and white males and females (Hughes, Cutter, Donahue, Friedman, Hulley, Hunkeler, Jacobs, Liu, Orden, Pirie, Tucker, and Wagenknecht 1987). This analysis is based on self-reported cigarette smoking status (yes/no) at the first four measurement occasions, or exams, from 1986 to 1993. The main focus of this article is on weighted generalized estimating equations (WGEE; Robins, Rotnitzky, and Zhao 1995) that are used in the estimation of smoking rates where observations or person-visits are weighted proportionally to their probability of being observed. The cluster-level weights of Fitzmaurice, Molenberghs, and Lipsitz (1995) were not considered as they may provide less efficient estimation than the observation-level weights considered here (O'Hara-Hines, Hines, and Friesen 2000). However, smoking rates are computed using a comparative approach based on maximum likelihood estimation due to Galecki, Ten Have, and Molenberghs (2000), who built on previous work by Molenberghs and Goetghebeur (1997).

In longitudinal studies, subjects commonly have missing data due to missed visits. A common pattern of missing data arises when subjects drop out of a study or are lost

to follow-up. A subject is called a dropout when the response variable is observed through a certain visit and is missing for all subsequent visits (Diggle, Liang, and Zeger 1994). This article is concerned both with dropouts and subjects with arbitrary or intermittent missing data patterns. The problem of missing data can be particularly acute in epidemiological cohort studies where subjects are followed prospectively over a period spanning several years. Possible assumptions about the *missingness mechanism* are that the data are missing completely at random (MCAR), missing at random (MAR), or informative (Little and Rubin 1987; Troxel, Lipsitz, and Brennan 1997). The data are MCAR if the missingness and measurement processes are independent. Under this scenario, the probability of missingness does not depend on the responses or covariates, whether observed or unobserved. The data are MAR if missingness depends on observed data but not on unobserved data. In repeated measures studies, an assumption of MAR depending on observed outcomes, which we refer to as MAR-Y, allows missingness to depend on the observed responses from previous visits as well as observed covariates. Conditional on these observables, missingness does not depend on the current or future responses. A MAR assumption of covariate-dependent missingness (MAR-X) allows the missingness of responses to depend only on observed covariates. Finally, missing data is informative or nonignorable (NI) if the missingness is related to the unobserved responses or covariates. The assumptions of MCAR or MAR are called *ignorable* because likelihood-based analyses are valid without statistical adjustments for the dropouts (Little 1995).

Inference for longitudinal binary outcomes often places primary interest in modeling marginal proportions, with parameters describing the association of an individual's repeated measures regarded as nuisance parameters. Broadly, estimation procedures include likelihood-based methods (e.g., Fitzmaurice and Laird 1993) and estimating equation methods. In particular, we consider the generalized estimating equations (GEE) approach of Liang and Zeger (1986)

John S. Preisser is Research Assistant Professor, Department of Biostatistics, University of North Carolina, Chapel Hill, NC 27599 (E-mail: jpreisse@bios.unc.edu). Andrzej T. Galecki is Assistant Research Scientist, Institute of Gerontology, University of Michigan, Ann Arbor, MI 48109. Kurt K. Lohman is Biostatistician, and Lynne E. Wagenknecht is Associate Professor of Epidemiology, Department of Public Health Sciences, Wake Forest University School of Medicine, Winston-Salem, NC 27157. The first author acknowledges support of grant IRG-198A from the American Cancer Society; the second author acknowledges support of NIA, NIH grant P30AG08808-05. The authors thank the editor and three anonymous referees for helpful comments.

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Table 1. Characteristics of Study Sample

Variable	Race/Sex group			
	Black males	Black females	White males	White females
Number of participants	1,145	1,473	1,160	1,300
Age at year 0 (mean years)	24.2	24.4	25.4	25.4
Attained education (%)				
High school or less	39.5	28.8	17.5	15.9
Some college	42.1	51.0	29.5	27.6
College degree	18.4	20.2	53.0	56.5
Percent smokers				
Year 0	36.9	31.3	26.5	27.3
Year 2	38.4	30.6	25.7	24.0
Year 5	37.2	31.9	24.3	21.9
Year 7	36.8	29.5	21.8	20.5

and Prentice (1988), who proposed correlation models for the association between pairs of responses on the same individual or cluster. Lipsitz, Laird, and Harrington (1991) and Fitzmaurice, Molenberghs, and Lipsitz (1995) proposed use of the odds ratio for binary data. Unlike likelihood-based analyses, the dropout pattern under a MAR-Y assumption in a GEE analysis cannot be ignored. In particular, the GEE approach will yield unbiased estimates of the regression parameters from the marginal mean model only if the data are MCAR or MAR-X. Robins et al. (1995) proposed WGEE and showed it to be valid under a MAR-Y assumption if the model for missingness is correctly specified. Heyting, Tolboom, and Essers (1992) proposed a similar weighting analysis for the problem of dropouts in clinical trials. They described a method for comparison across groups of a final endpoint, whereas the method of Robins et al. (1995) is of such generality to be applicable to the analysis of trends. A data imputation approach when data are MAR-Y was proposed by Paik (1997), but is not considered here.

The WGEE approach is compared to a likelihood approach through an application to data on smoking trends from the CARDIA study. Section 2 describes the data and model used to estimate 7-year trends in smoking prevalence among the four race/sex groups. Section 3 reviews the WGEE approach and describes a WGEE analysis of the smoking data modified such that visits are missing only due to dropouts. A test related to that of Diggle (1989) rejects the covariate-dependent missingness assumption, invalidating the GEE approach. Sections 4 and 5 illustrate WGEE and a likelihood-based approach for intermittent missing data patterns by analyzing all available data including visits observed after missed visits. Smoking trend estimates are adjusted for U.S. Census age and education distributions in section 6. Finally, section 7 summarizes statistical approaches to incomplete longitudinal binary data.

2. SMOKING TRENDS AMONG YOUNG ADULTS

In the CARDIA study conducted in four urban centers (Birmingham, AL; Chicago, IL; Minneapolis, MN; and Oakland, CA) across the United States, a total of 5,115 young adults ages 18–30 years were followed prospectively and examined up to five times from 1986 to 1996. Recruitment, restricted to blacks and whites, was carried out to

achieve approximate balance in sample size with respect to age, race, gender, and education. Within targeted subgroups, participants were generally recruited by telephone, from selected census tracts, or from a health plan membership (Hughes et al. 1987). Study participants were scheduled for visits at years 0, 2, 5, 7, and 10. We consider the first four visits and the 5,078 (99.3%) young adults with self-reported smoking status (yes/no) known at baseline (year 0). Table 1 shows that, on average, young adults in the four groups were similar with respect to age, and that whites were more likely to have a higher level of attained education (at study end) than blacks. Furthermore, CARDIA participants are considerably more educated than young adults in the general U.S. population. Of CARDIA participants born 1963–1967, 54.8% of whites and 19.4% of blacks attained at least a bachelor's degree. The corresponding 1992 U.S. Census Population survey rates for white and black young adults ages 25–29 are 25.0% and 11.3% (U.S. Bureau of the Census, 1998a). Finally, Table 1 shows that 7-year changes in smoking prevalence for the sample are -1.1 , -1.8 , -4.7 , and -6.8 in percentage points for black males, black females, white males, and white females, respectively, so that the observed smoking rates decreased over time for each group, although only slightly for blacks.

The goal of the statistical analysis is to draw inferences on the change in smoking prevalence by race/sex group in the presence of missing data and intraperson correlation. Let $Y_{ijt} = 1$ if the i th person in the j th race/sex group is a smoker at the t th visit, and 0 if the participant is a nonsmoker. The model of interest for the probability that the i th person, $i = 1, \dots, 5,078$, is a smoker at time t is

$$\text{logit}(\pi_{ijt}) = \text{logit}[P(Y_{ijt} = 1)] = \beta_{0j} + \beta_{1j}x_t \quad (1)$$

for $j = 1, 2, 3, 4$, and years since the baseline measurement $x_t = 0, 2, 5, 7$ for $t = 1, 2, 3, 4$, respectively. Although GEE is commonly used to estimate such regression parameters, inspection of the data suggests that the covariate-dependent missingness assumption of GEE may not be satisfied for the smoking data. In particular, Table 2 shows that participants who were smokers at baseline were more likely to have subsequent missing visits than participants who were nonsmokers at baseline. Also note that blacks are more likely to have missing visits than whites.

Table 2. Occasion-Specific Sample Sizes by Baseline Smoking Status

Baseline smoker	Year exam	Black males	Black females	White males	White females
No	0	722	1,012	853	945
	2	616 (85%)	906 (90%)	806 (95%)	901 (95%)
	5	581 (81%)	835 (83%)	783 (92%)	868 (92%)
	7	535 (74%)	790 (78%)	752 (88%)	816 (86%)
Yes	0	423	461	307	355
	2	356 (84%)	374 (81%)	285 (93%)	320 (90%)
	5	315 (75%)	370 (80%)	260 (85%)	299 (84%)
	7	285 (67%)	338 (73%)	246 (80%)	280 (79%)

3. WEIGHTED GENERALIZED ESTIMATING EQUATIONS ANALYSIS OF LONGITUDINAL DATA WITH DROPOUTS

Now, we consider statistical analyses for the smoking data that require patterns of missing data that arise only from dropouts. The advantage of this approach is that assumptions underlying a model for dropouts are weaker than assumptions required for a model that allows intermittent missing data patterns (Robins et al. 1995). The disadvantage is potential loss of efficiency when data from person-exams occurring after a person’s first missed exam are omitted in order to create a dataset with only monotone missingness patterns (Table 3). For the smoking data, 578 person-exams out of a total of 17,995 were omitted to create a monotone data set. The resulting 7-year follow-up rates for the monotone data set were 62%, 68%, 81%, and 79% for black males, black females, white males, and white females, respectively.

3.1 Testing the Covariate-Dependent Missingness Assumption

The assumption of covariate-dependent missingness or MAR-X for dropouts may be tested using a two-step procedure. The first step involves constructing separate tests for each combination of time, $t = 2, 3, 4$, and race/sex group. Define the outcome variable $R_{it} = 1$ if the smoking status of the i th individual at the t th visit is observed and

$R_{it} = 0$ if it is missing. Maximum likelihood is used to fit separate logistic regression models for each group $\times$ time combination with the following covariates: smoking at time $t - 1$, age, and attained education. If for each combination, dropouts are MAR-X, depending only on age (in years) and attained education, then the probability of dropout does not depend on smoking at time $t - 1$. The second step of the procedure applies the Kolmogorov–Smirnov test to evaluate whether the p values corresponding to Wald tests for the effect of smoking at previous visit behave like a random sample from a uniform $U(0, 1)$ distribution. The implicit assumption that the 12 tests are independent follows from the fact that once an individual drops out, he or she is no longer available for subsequent testing. Rejecting the null hypothesis that the p -values follow a uniform distribution corresponds to rejecting MAR-X in favor of MAR-Y. A probability plot may accompany the test for descriptive purposes; the greater the deviation of the points from the 45° line, the greater the evidence against MAR-X. The two-step procedure is described using nonparametric tests in the first step by Diggle (1989) and Diggle et al. (1994). For the CARDIA data, the 12 resulting p -values, $p_u, u = 1, \dots, 12$, based on Wald tests for smoking at time $t - 1$ are shown in the probability plot in Figure 1. The one-sided Kolmogorov–Smirnov test statistic, $D_+ = \sup\{\hat{F}(p_u) - p_u\} = .342$, where $\hat{F}(p_u) = u/12$, gives an asymptotic-based p -value = .023, rejecting MAR-X; the value of D_+ observed is the vertical distance from the circled p -value and the 45° line in Figure 1. Each of the four race/sex groups has a single small p_u (less than .07) suggesting a MAR-Y effect; three corresponded to year 5, and one to year 2. An alternative test for MAR-X based on the binomial $B(12, .05)$ distribution gives a p -value of .02, which is equal to the probability of observing 3 or more p -values out of 12 that are less than .05.

3.2 Weighted Estimating Equations for Longitudinal Data

Some notation is required for the presentation of the WGEE approach. Assume a set of T observation times common to all individuals, and define the complete data response vector $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{iT})'$ for individual $i, i = 1, \dots, K$. For example, if a subject possesses the outcome of interest at time t (subject is a smoker) $Y_{it} = 1$, and $Y_{it} = 0$ if the subject does not possess this characteristic (subject is not a smoker). The corresponding complete covariate matrix for individual i is $\mathbf{X}_i = (\mathbf{X}_{i1}, \dots, \mathbf{X}_{iT})'$ where $\mathbf{X}_{it}$ is a $p \times 1$ covariate vector at time t .

Now, suppose the data follow monotone patterns of missingness such that individual i is observed only at times $t = 1, \dots, T_i$, giving the $T_i \times 1$ vector of observed responses, $\mathbf{Y}_i^o = (Y_{i1}, \dots, Y_{iT_i})', 1 \leq T_i \leq T$. Likewise, let $\mathbf{X}_i^o = (\mathbf{X}_{i1}, \dots, \mathbf{X}_{iT_i})'$ represent the corresponding matrix of covariates for the observed time points. Next, let $E(Y_{it}|\mathbf{X}_{it}) = \pi_{it}$ be the probability that $Y_{it} = 1$ given $\mathbf{X}_{it}$, and let $\boldsymbol{\pi}_i = (\pi_{i1}, \dots, \pi_{iT_i})'$ and $\boldsymbol{\pi}_i^o = (\pi_{i1}, \dots, \pi_{iT_i})'$. We consider marginal models for longitudinal studies that relate the expected value of an individual’s binary response at time t (e.g., the probability of being a smoker at time

Table 3. Data on Cigarette Smoking From 1986–1993 for 5,078 Young Adults From the CARDIA Study

	$Y_1Y_2Y_3Y_4$	Black males	Black females	White males	White females
pppp	0000	402	634	660	709
	0001	11	7	11	10
	0010	8	7	8	8
	0011	14	13	6	12
	0100	11	14	8	18
	0101	3	2	1	3
	0110	1	6	6	4
	0111	22	32	16	11
	1000	14	14	15	35
	1001	1	2	2	5
	1010	2	8	2	7
	1011	12	16	13	14
	1100	10	20	22	20
	1101	7	7	5	13
	1110	9	11	17	29
	1111	188	213	147	135
pppa	000a	49	73	39	62
	001a	3	4	0	0
	010a	1	1	0	4
	011a	8	4	5	1
	100a	3	2	4	4
	101a	2	0	3	1
	110a	8	2	1	1
	111a	36	31	17	20
ppaa	00aa	71	100	45	56
	01aa	12	9	1	3
	10aa	7	9	3	5
	11aa	57	39	34	31
paaa	0aaa	106	106	47	44
	1aaa	67	87	22	35

NOTE: a = absent, p = present; this is the monotone data set created by omitting visits that occurred after a person’s first missed visit.

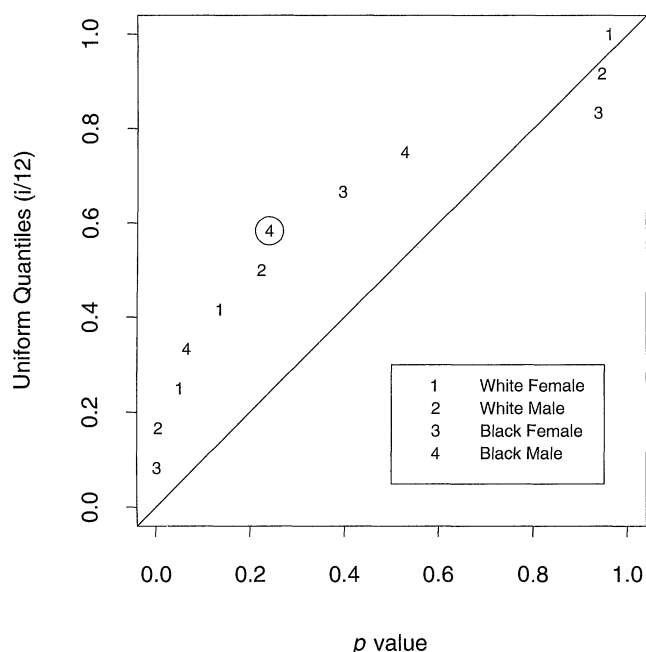


Figure 1. Probability Plot for Test of MAR-X. The circled p -value is the one giving the maximum vertical distance to the line, the Kolmogorov-Smirnov test statistic.

t) to covariates via a known link function $g(\cdot)$ (McCullagh and Nelder 1989). Our goal is to make inferences about marginal probabilities, $\pi_{it}(\beta)$, where $g(\pi_{it}) = \mathbf{X}_{it}'\beta$, for individuals regardless of whether or not they drop out. We consider generalized linear models for π_{it} further specified by $\text{var}(Y_{it}) = v(\pi_{it})\phi$ for some variance function $v(\cdot)$ and scale parameter, ϕ . For the binary smoking data, $\beta' = (\beta_{01}, \beta_{02}, \beta_{03}, \beta_{04}, \beta_{11}, \beta_{12}, \beta_{13}, \beta_{14})$ in model (1), $v(\pi_{it}) = \pi_{it}(1 - \pi_{it})$, and $\phi = 1$.

The conventional approach to estimation of regression parameters in models for marginal means such as (1) is the GEE procedure (Diggle et al. 1994). To estimate β , while accounting for correlation among an individual's repeated responses, Liang and Zeger (1986) proposed generalized estimating equations,

$$\sum_{i=1}^K \mathbf{D}_i'(\mathbf{X}_i^0, \beta)(\mathbf{V}_i^0)^{-1}[\mathbf{Y}_i^0 - \boldsymbol{\pi}_i^0(\beta)] = 0. \quad (2)$$

where $\mathbf{D}_i'(\mathbf{X}_i^0, \beta) = \partial \boldsymbol{\pi}_i^0 / \partial \beta$ is a $T_i \times p$ matrix and $\mathbf{V}_i^0$ is a working covariance matrix for $\mathbf{Y}_i^0$. A working model for the correlation is assumed such that $\mathbf{V}_i^0 = \mathbf{A}_i^0 \mathbf{C}_i^0 \mathbf{A}_i^0$, $\mathbf{A}_i^0 = \text{diag}\{v_{it}^{1/2}\}$ is a $T_i \times T_i$ diagonal matrix, where $v_{it} = v(\pi_{it})$ and $\mathbf{C}_i^0(\rho)$ is a working correlation matrix that depends on an unknown nuisance parameter vector ρ . Equation (2) yields a consistent estimate of β if the data are MCAR or MAR-X, whereas, under MAR-Y, a GEE analysis may give biased estimates.

The rejection of MAR-X in the previous section indicates that GEE is not a valid approach for estimation of (1). By contrast, the WGEE approach is valid under a MAR-Y (or more generally, MAR) assumption even if the correlation model is misspecified, provided the model for estimating the probability for missing response is correctly specified. Applying WGEE involves (i) determining weights through

a model for the missingness, (ii) applying the weights in the estimation of parameters in the marginal mean model (1), and (iii) calculating the empirical sandwich-type variance estimator with the estimation of weights taken into consideration.

To characterize the missing data process, let $\lambda_{it} = P(R_{it} = 1 | R_{i(t-1)} = 1, \mathbf{Z}_i, \alpha)$ denote the probability of observing a response at time t for the i th individual, conditional on the individual being observed at the time $t - 1$. Let $\mathbf{Z}_i = (\mathbf{X}_i^0, \mathbf{X}_i^{(a)}, \mathbf{Y}_i^0)$ where the missing data process may depend not only on observed responses and covariates in model (1) but also on additional covariates $\mathbf{X}_i^{(a)}$ distinct from $\mathbf{X}_i^0$. Here, we do not consider intermittent missing data patterns (i.e., $R_{it} = 0, R_{i(t+k)} = 1$, for $k > 0$ is not allowed). For the first time point, assume $R_{i1} = 1$ and define $\lambda_{i1} = 1$. The MAR assumption of WGEE specifies that $\lambda_{it} = P(R_{it} = 1 | R_{i(t-1)} = 1, \mathbf{X}_i^0, \mathbf{X}_i^{(a)}, Y_{i1}, \dots, Y_{i(t-1)}, \alpha)$, so that conditional on observed covariates and past responses $(Y_{i1}, \dots, Y_{i(t-1)})$, R_{it} and Y_{it} are independent. The missing data mechanism therefore depends only on observed data and is specified up to a $q \times 1$ vector of unknown parameters, α . We obtain $\hat{\lambda}_{it}$ by fitting a logistic model, $\text{logit}\{\lambda_{it}(\alpha)\} = \mathbf{Z}_{it}\alpha$, for $t = 2, \dots, T$. The log-partial likelihood for the i th subject is

$$\sum_t R_{i,t-1} \log\{\lambda_{it}(\alpha)^{R_{it}} [1 - \lambda_{it}(\alpha)]^{1-R_{it}}\}.$$

Differentiation with respect to α yields $\mathbf{S}_i(\alpha) = \sum_t R_{i,t-1} \mathbf{Z}_{it} [R_{it} - \lambda_{it}(\alpha)]$, the α -score component of the i th individual. The summation in the i th score component arises from repeated conditioning based on the conditional independence of the $R_{it}, t = 2, \dots, T$. The partial likelihood estimate $\hat{\alpha}$ and thus $\hat{\lambda}_{it}$'s are obtained by solving the estimating equations $\sum_i \mathbf{S}_i(\alpha) = 0$. The weight, w_{it} , for the i th individual at the t th time is the inverse of the unconditional probability of being observed at time t , estimated as the inverse of the cumulative product of conditional probabilities, $\hat{w}_{it}^{-1} = \hat{\lambda}_{i1} \times \dots \times \hat{\lambda}_{it}$. An observation with a low probability of being observed will receive a large weight.

Once weights are obtained, a consistent estimate of β may be obtained from

$$\sum_{i=1}^K \mathbf{D}_i'(\mathbf{X}_i, \beta) \mathbf{V}_i^{-1} \mathbf{W}_i [\mathbf{Y}_i - \boldsymbol{\pi}_i(\beta)] = 0, \quad (3)$$

where $\mathbf{D}_i(\mathbf{X}_i, \beta) = \partial \boldsymbol{\pi}_i / \partial \beta$ and $\mathbf{V}_i = \mathbf{A}_i \mathbf{C}_i \mathbf{A}_i$ is a $T \times T$ working covariate matrix for $\mathbf{Y}_i$ (Robins et al. 1995). Here, the dropout process is taken into account through specification of a $T \times T$ diagonal matrix of occasion-specific weights, $\mathbf{W}_i = \text{diag}\{R_{i1}w_{i1}, \dots, R_{iT}w_{iT}\}$, where $R_{it} = 1$ if the i th individual's response is observed at time t , and 0 otherwise. In other words, the weight is given by w_{it} for observed visits and 0 for unobserved visits. As in GEE, iteratively reweighted estimation of β alternates at each step with method of moments estimation of ρ . Under correctly specified models for the marginal means and for the dropout process, equation (3) yields a consistent estimate of β , which has an asymptotic normal distribution with con-

sistent estimator of its asymptotic variance given by

$$\left(\sum_{i=1}^K \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{W}_i \mathbf{D}_i \right)^{-T} \times \sum_{i=1}^K \mathbf{E}_i \mathbf{E}_i' \left(\sum_{i=1}^K \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{W}_i \mathbf{D}_i \right)^{-1}, \quad (4)$$

where $\mathbf{E}_i = \mathbf{U}_i - (\sum_{i=1}^K \mathbf{U}_i \mathbf{S}_i') (\sum_{i=1}^K \mathbf{S}_i \mathbf{S}_i')^{-1} \mathbf{S}_i$, $\mathbf{U}_i = \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{W}_i (\mathbf{Y}_i - \pi_i)$, and $\mathbf{S}_i = \mathbf{S}_i(\alpha)$. Note that $\sum_{i=1}^K \mathbf{E}_i \mathbf{E}_i'$ is the matrix sum of squares and cross-products of the residuals from the multivariate regression of the quasi-score component from (3) on the score vectors from the dropout model. The use of $\sum_{i=1}^K \mathbf{E}_i \mathbf{E}_i'$ in the center of (4) instead of $\sum_{i=1}^K \mathbf{U}_i \mathbf{U}_i'$ adjusts for estimation of α . Efficiency of the WGEE estimates from (3) increases (or at least does not diminish) when more predictor variables are added to the missing data model (Robins et al. 1995).

Note that unless $\mathbf{C}_i = \mathbf{I}$, (3) depends on covariates from both observed and unobserved occasions. For an individual who drops out of the study prior to time T , $\mathbf{X}_i$ will be known, as is required for (3), if all covariates are baseline characteristics (time-independent) or nonstochastic functions of known quantities such as time or polynomials of time. The choice of the working correlation matrix in (3) may affect efficiency (Troxel 1998). Like GEE, the use of sandwich variance estimators in WGEE provides robustness, in an asymptotic sense, to misspecification of the correlation structure. With consistent estimation of weights provided by a correctly specified dropout model, WGEE does not require correct specification of the correlation structure in order to estimate β and the variance of its estimate consistently.

One concern with the WGEE method is that misspecifying the missingness model may cause bias in $\hat{\beta}$. As illustrated by the simulation experiment of Robins et al. (1995), particular problems may arise when some observations receive inappropriately large weights corresponding to very low predicted probabilities of being observed. Because individuals with large weights will have a large influence on the analysis, one should judge whether such weights are appropriate. A simple diagnostic procedure is to check the

largest weight ($\max\{w_{iT}\}$) that is the inverse of the minimum (over all participants) of the estimated probability of remaining in the study. A large $\max\{w_{iT}\}$ need not imply problems with the analysis if further scrutiny of data points with large weights supports the appropriateness of the missingness model.

3.3 Analysis of Smoking Trends With Dropouts Present

We now apply WGEE to the estimation of trends in cigarette smoking in the CARDIA study. The first step is to determine weights by identifying a missing data model. Table 4 reports deviances for selected nested models, labeled A through H, fitted to the monotone data set. The degrees of freedom refer to the number of additional terms in the model in the next line, and Δ gives the difference in the deviance between the two models. Nested models may be compared because Δ has an asymptotic chi-square distribution when the smaller model is correct. A comparison of models D and F gives $\Delta = 21.0$ with 4 degrees of freedom providing strong support for MAR-Y that further depends on race and/or education. Because additional two-way interaction terms are not significant, model F provides a good fit to the data. However, our goal is not to identify a parsimonious missing data model nor make inference on α , but to identify a model that provides stable weights for the estimation of smoking trends. Weights are considered stable if they change very little when additional terms are added to the missingness model.

With these considerations, the analysis used dropout model G. Results in Table 5 suggest that the magnitude of the MAR-Y effect varies by level of attained education. Young adults with attained college degree are estimated to have $1/\exp(-.67) = 1.95$ times higher odds to miss a visit if they were a smoker (vs. nonsmoker) at the previous visit; there is very little evidence of a MAR-Y effect for young adults with less than a college degree. Table 6 gives estimates of smoking trends, β_{1j} in (1); quadratic effects of time were not significant. Even though dropout model G did not have any statistically significant terms beyond model F, there is a notable difference in the values of WGEE estimated slopes for black males between the two dropout models. Conversely, the use of model G gives WGEE estimates similar to those based on model H that has all two-way

Table 4. Deviances of Nested Logit Models for the Missingness Process

Model	Additional variables	Degrees of freedom (df)	Monotone		Full	
			Deviance	Δ	Deviance	Δ
A	$b, m, b \times m$	3	8,828.45	160.15	12,709.44	364.69
B	a, e	2	8,668.30	2.04	12,344.75	220.73
C	t	2	8,666.26	9.79	12,124.02	4.10
D	$b \times t$	1	8,656.47	11.55	12,119.92	12.51
E	y_{t-1}	3	8,644.92	9.45	12,107.41	20.2
F	$b \times y_{t-1}, e \times y_{t-1}$	4	8,635.47	1.98	12,087.21	1.61
G	$m \times y_{t-1}, a \times y_{t-1}, t \times y_{t-1}$	8	8,633.49	2.75	12,085.60	8.89
H	$m \times t, a \times b, a \times m, e \times b, e \times m$	1	8,630.74	—	12,076.71	1,576.04
I	r_{t-1}	1	—	—	10,500.67	3.1
J	$r_{t-1} \times y_{t-1}$	—	—	—	10,497.57	—

NOTE: Δ is the change in deviance with respect to the model in the line below. The variables are defined as follows: b = black, m = male, a = age, e = attained education, t = year, y_{t-1} = smoking status at previous visit for monotone data set, and smoking status at most recent prior visit for intermittent dataset; r_{t-1} = missingness status at previous visit.

Table 5. Logit Model Parameter Estimates for Missingness Model G

Variable	Monotone	Full data set
Intercept	2.22 (.28)	2.30 (.23)
Black race	-.59 (.13)	-.58 (.12)
Male sex	.10 (.10)	.07 (.09)
Black × male	-.26 (.12)	-.24 (.10)
Age (in 10 year units)	.39 (.10)	.40 (.08)
High school (HS) or less	-.91 (.10)	-1.14 (.08)
Some college	-.51 (.09)	-.55 (.08)
Year 5	-.33 (.13)	-.62 (.12)
Year 7	-.37 (.13)	-1.08 (.11)
Black × year 5	.31 (.14)	.16 (.13)
Black × year 7	.46 (.15)	.27 (.13)
Smoking	-.67 (.45)	-.81 (.37)
Smoking × black	.07 (.13)	.11 (.11)
Smoking × male	.02 (.12)	.09 (.10)
Smoking × age (10 years)	-.04 (.16)	-.01 (.13)
Smoking × HS or less	.44 (.18)	.52 (.15)
Smoking × some college	.50 (.18)	.59 (.15)
Smoking × year 5	.09 (.14)	.08 (.13)
Smoking × year 7	.21 (.15)	.11 (.12)

NOTE: Smoking is at previous visit for monotone data and most recently observed prior visit for intermittent data; college degree and year 2 are reference categories for attained education and year of follow-up, respectively.

interactions. Conclusions with WGEE are similar for different working correlation structures. There is a nonsignificant increase in smoking trends among black males, very little change among black females, and statistically significant decreases in smoking among white males and females. The largest $\max\{w_{i4}\}$ for any of the four missingness models shown in Table 6 was 2.39. Furthermore, this value exceeded that of any other missingness model given in this paper so that no CARDIA participant had a large influence on any of the WGEE analyses.

Three final comments are made regarding Table 6. First, the standard errors of the estimates for slopes describing trends in smoking based on weights from model G are smaller than those based on the simpler model F, illustrating the theory of Robins et al. (1995). Second, ignoring the MAR-Y effect, GEE tended to give inappropriately smaller slope estimates relative to WGEE, particularly when the working independence correlation structure was used. As smokers are more likely to drop out, the independence estimating equations give too much relative weight to the non-smoker observations at later time periods relative to smokers. Third, working independence and exchangeable correlation structures gave similar WGEE results (equal to two decimal places in Table 6); generally, when all covariates are cluster level, the two sets of estimating equations are equivalent. We favor results from the unstructured correlation matrix, because estimated correlations among points farther apart in time were less correlated than pairs of points closer together; correlation estimates among pairs of times ranged from .71 (between exams 1 and 4) to .88 (between exams 3 and 4).

4. WEIGHTED GENERALIZED ESTIMATING EQUATIONS ANALYSIS WITH INTERMITTENT MISSING DATA PATTERNS

Additional assumptions about the missing data mechanism are required to apply WGEE to the full dataset, which

contains subjects with intermittent patterns of missing data. Extensions of the methodology reviewed in section 3.2 involve conditioning on the past history that now includes observed and unobserved visits. As fitting the missing data model involves using only observed data, the additional assumption required is that the probability of missingness at a current visit is independent of unobserved past smoking history, given observed past smoking history and covariates. Because this assumption is stronger than that for monotone data given in section 3.2, WGEE analysis of the smoking data with intermittent missing data patterns plays a supporting role to the primary analysis based on the monotone estimator.

To allow some generality for intermittent missing data, let the probability of missingness at time t further depend on an individual's response history $\bar{\mathbf{R}}_{it} = (R_{i1}, \dots, R_{i(t-1)})$. Redefine, $\lambda_{it} = P(R_{it} = 1 | \bar{\mathbf{R}}_{it}, \mathbf{Z}_i, \alpha)$ as the probability of observing a response at time t for the i th individual conditional on that individual's missingness history, $\bar{\mathbf{R}}_{it}$, and their observed data, $\mathbf{Z}_i$. Recall that the observation level weight in WGEE is the inverse of the unconditional probability that the i th individual is observed at time t . This marginal probability is the sum of the joint probabilities of the different response profiles or paths that an individual may take to be observed at time t . For monotone data, there is only one path, expressed by the marginal (and joint) probability $P(\bar{\mathbf{R}}_{it} = \mathbf{1}, R_{it} = 1)$. For intermittent missing data, we assume $P(R_{i1} = 1) = 1$ and further that $P(R_{it} = 1 | \bar{\mathbf{R}}_{it}, \mathbf{Z}_i, \alpha) = P(R_{it} = 1 | R_{i(t-1)}, \mathbf{Z}_i, \alpha)$, so that conditional on $\mathbf{Z}_i$ and whether the subject missed the previous visit, the probability of missing the current visit is independent of missingness history prior to time $t - 1$. For fixed $\mathbf{Z}_i$ define $\lambda_{it}^{(j)} = P(R_{it} = 1 | R_{i(t-1)} = j, \mathbf{Z}_i, \alpha)$, $j = 0, 1$, for $t = 3, \dots, T$. The marginal probabilities of being observed (inverse weights) are $w_{i1}^{-1} = 1$, $w_{i2}^{-1} = \lambda_{i2}$, and $w_{it}^{-1} = \lambda_{it}^{(1)} w_{i(t-1)}^{-1} + \lambda_{it}^{(0)} (1 - w_{i(t-1)}^{-1})$, $t = 3, \dots, T$. Note that if, conditional on $\mathbf{Z}_i$, missingness does not depend on past missingness history, that is, $\lambda_{it} = \lambda_{it}^{(j)}$, $j = 0, 1$, then the weights reduce to $w_{it}^{-1} = \lambda_{it}$. These more simple weights will be applied in the next section. Note that they do not apply when data are restricted to monotone patterns of missingness, because in that case, the probability

Table 6. Estimates of β_{ij} ($\times 100$) and Standard Errors for Monotone Data Set

Model	Black males	Black females	White males	White females
GEE-Ind	-.63 (.93)	-1.21 (.80)	-3.79 (.89)	-5.99 (.92)
GEE-Exch	.88 (.74)	-.20 (.64)	-2.10 (.74)	-4.48 (.78)
GEE-Unst	.89 (.74)	-.21 (.64)	-2.12 (.74)	-4.55 (.78)
WGEE(E)-Exch	1.32 (.89)	.40 (.76)	-2.41 (.86)	-4.51 (.90)
WGEE(F)-Exch	1.04 (.86)	.13 (.74)	-2.25 (.83)	-4.29 (.87)
WGEE(G)-Exch	.89 (.81)	.11 (.69)	-2.31 (.81)	-4.28 (.85)
WGEE(H)-Exch	.89 (.81)	.12 (.69)	-2.35 (.81)	-4.28 (.85)
WGEE(E)-Unst	1.25 (.90)	.21 (.77)	-2.59 (.88)	-4.44 (.93)
WGEE(F)-Unst	.96 (.87)	-.06 (.74)	-2.43 (.85)	-4.23 (.90)
WGEE(G)-Unst	.77 (.82)	-.10 (.69)	-2.51 (.82)	-4.25 (.88)
WGEE(H)-Unst	.76 (.81)	-.10 (.69)	-2.54 (.83)	-4.24 (.87)

NOTE: Empirical standard errors are reported for GEE and WGEE. WGEE results for independence were the same to two decimal places as for exchangeable.

of missingness depends on past missingness history because $P(R_{it} = 1 | \bar{R}_{it} \neq 1) = 0$.

4.1 Analysis of Smoking Trends Using all Available Data

To apply WGEE to the full dataset that includes intermittent missing data patterns, we must first choose a missingness model from which to estimate weights. We consider models analogous to the monotone case, but include the smoking status at the most recently observed prior visit (for models E through J in Table 4) instead of smoking status at the previous visit. Several features in Table 4 are particularly striking. First, the deviances from the full model are considerably larger than the deviances from the monotone model because more observations were used to fit the missingness model in the intermittent data case. We modelled the probability of being observed using outcomes from years 2, 5, and 7 for every study participant in accord with the philosophy that no individual is a dropout; rather, each has some nonzero probability of being observed at any exam. Because the individuals in CARDIA are young adults, the probability of loss to follow-up due to death is rare, so the modelling strategy is sensible. Next, Table 4 data show a very large drop in deviance when a term for missingness status at the previous visit $r_{i(t-1)}$ was included, as compared to a small change due to further adding the interaction term of previous missingness status by most recently observed prior smoking status. In missingness model J, parameter estimates (standard errors) for the additional terms were 2.795 (.091) and $-.256 (.145)$, respectively. Therefore, among persons who were nonsmokers at the most recently observed prior exam, a person who was observed at the previous time point has an estimated 16.4 times higher odds of being observed at the current time point than a person who was not observed at the previous time point. Similarly, among persons who were smokers at the most recently observed prior exam, a person who was observed at the previous time point has an estimated 12.7 times higher odds of being observed at the current time point than a person who was not observed at the previous time point. The interaction term was not significant at the .05 significance level, suggesting that the assumption of a common odds ratio for smokers and nonsmokers in model I

is reasonable. For relative computational simplicity as well as comparability to the analysis for monotone data (Table 5), our analysis is based on missingness model G. Justification for not including $r_{i(t-1)}$ in the model is given next. The WGEE results shown in Table 7 are qualitatively similar to their monotone data counterparts in Table 6. There was a nonsignificant increase in smoking among black males, very little change for black females, and significant decrease in smoking among both white males and white females. We obtained similar results when baseline smoking status y_{i1} was included in the missingness model instead of most recently observed prior smoking status.

4.2 A Small Simulation Study

The previous analysis assumes that ignoring the strong effect of previous missingness status on current missingness status has very little effect on the estimation of the β_{1j} . To explore this, we conducted a small simulation study where interest is in the slope parameter, β_1 , for a single group. In the simulations, we set $T = 3$, generating three correlated binary responses for each individual, $\mathbf{Y}_i = (Y_{i1}, Y_{i2}, Y_{i3})$, indicating smoking status at three points in time. We assumed the marginal probability of being a smoker at time $t = 1, 2, 3$ is specified correctly as $\text{logit}[P(Y_{it} = 1)] = \beta_0 + \beta_1(t - 1)$. We generated data, $\mathbf{Y}_i$, by fixing $\beta_0 = -.5$ and $\beta_1 = .1$ corresponding to marginal probabilities $\pi_i = (.378, .401, .426)$ indicating a moderate increase in smoking prevalence over time. To complete the description of the joint distribution, we define $\rho = (\rho_{12}, \rho_{13}, \rho_{23}, \rho_{123})$ where $\rho_{tt'} = \rho_{itt'}$ is the correlation between Y_{it} and $Y_{it'}$ assumed constant across individuals, and ρ_{123} is a constant third-order association parameter (Lipsitz, Fitzmaurice, Orav, and Laird 1994). We chose $\rho = (.6, .6, .6, .3)$ to reflect the strong intraperson correlation for being a smoker in the CARDIA study, and to satisfy the natural restrictions of the cell probabilities. We fixed $R_{i1} = 1$ and generated missing data indicators R_{i2} and R_{i3} based on missingness models of the form

$$\text{logit}(\lambda_{it}) = \alpha_0 + \alpha_1 I_{(t=3)} + \alpha_2 y_i^* + m_{i(t-1)}(\alpha_3 + \alpha_4 y_i^*), \quad (5)$$

for $t = 2, 3$, and where $I_{(t=3)} = 1$ if $t = 3$ and 0 otherwise; $y_i^* = 2y_{i1} - 1$ and $m_{i(t-1)} = 1 - 2r_{i(t-1)}$ are indicators of baseline smoking and previous missingness status, respectively. Observed data, $\mathbf{Y}_i^o$, were generated based on R_{i2} and R_{i3} , allowing the intermittent data possibility that $R_{i2} = 0$ and $R_{i3} = 1$. We set $\alpha_1 = -0.1$ throughout, and α_0 to 0 or 1.0. We generated R_{i2} and R_{i3} for 24 sets of simulations corresponding to all possible combinations of $\alpha_2 = \{0, -.1, -.2\}$, $\alpha_3 = \{0, -1.4\}$, and $\alpha_4 = \{0, .1, .5, 1.0\}$. When $\alpha_3 = 0$, we set $\alpha_0 = 1.0$, and when $\alpha_3 = -1.4$, $\alpha_0 = 0$. The effect was to keep the average percentage of missed visits from one set of simulations to another relatively constant, the point being to study the effect of different missing data mechanisms with the overall amount of missing data approximately constant. The particular combination $\alpha = (0, -.1, -.1, -1.4, .1)$ corresponds roughly to estimates obtained by applying model

Table 7. Estimates of $\beta_{1j} \times 100$ and Standard Errors for Full Data Set

Model	Black males	Black females	White males	White females
GEE-Ind	-.25 (.84)	-.65 (.72)	-3.37 (.83)	-5.24 (.86)
GEE-Exch	1.09 (.71)	-.26 (.61)	-2.19 (.73)	-4.11 (.75)
GEE-Unst	1.08 (.71)	-.31 (.61)	-2.23 (.73)	-4.21 (.74)
WGEE-Ind	.47 (.75)	.17 (.63)	-2.31 (.78)	-3.91 (.80)
WGEE-Exch	.47 (.75)	.17 (.63)	-2.31 (.78)	-3.91 (.80)
WGEE-Unst	.44 (.76)	-.06 (.64)	-2.45 (.79)	-3.85 (.82)
ML-Ind	-.25 (1.24)	-.65 (1.12)	-3.37 (1.32)	-5.24 (1.28)
ML-Exch	1.29 (.59)	-.31 (.53)	-2.53 (.65)	-3.77 (.64)
ML-Unst	1.09 (.68)	-.44 (.62)	-2.52 (.75)	-3.99 (.74)

NOTE: Empirical standard errors are reported for GEE and WGEE. WGEE results are based on weights from missingness model G that includes as predictor the most recently observed prior smoking status.

(5), generalized to $T = 4$, to the CARDIA data. For each of the 24 ways that data were generated, we applied WGEE with exchangeable working correlation structure, using weights $w_{it} = 1/\lambda_{it}$, determined from the missing data logit model that only included terms corresponding to regression coefficients α_0, α_1 , and α_2 . Each set of simulations consisted of 1,000 replicate runs each based on $K = 500$ subjects. Percent relative bias was computed as $(1/1000) \sum_{s=1}^{1000} (\hat{\beta}_s - \beta_1)/\beta_1 \times 100\%$ where $\hat{\beta}_s$ is the estimate of β_1 from the s th simulated replicate. Test size was defined as the percent of 95% Wald-type confidence intervals for β_1 using (3) and (4), respectively, that did not contain zero when simulations were run with $\beta_1 = 0$.

The average percent of visits that were missing in the simulated data ranged from 16.3% to 18.6% across the 24 sets of simulations. In CARDIA, 2,317 out of 20,312 (11.4%) of all visits were missing. Table 8 reports estimated percent relative bias for the different missingness mechanisms. The first cell of the table corresponds to $\alpha = (1, -.1, 0, 0, 0)$ or covariate-dependent missingness, and the entries in the last two columns in the first row correspond to MAR missingness patterns. As expected, there is negligible estimated bias shown in row 1 because the model for the missing data is correctly specified. Rows 2–8 show simulation results for misspecified missingness models. In row 2, the MAR effect does not depend on $R_{i(t-1)}$, and despite R_{i2} and R_{i3} being strongly correlated, percent relative bias is negligible. Furthermore, when $\alpha_4 = .1$ reflecting mildly different MAR effects based on previous missingness status (as found in the CARDIA study), the estimated bias is still very small (rows 3 and 4 of Table 8). When the interaction effect is stronger given by $\alpha_4 = .5$ (rows 5 and 6), the estimated bias is larger, and when $\alpha_4 = 1$ (rows 7 and 8), the bias is the largest for the scenarios considered. Finally, test size was near the nominal .05 level for rows 1–4 (range .048 to .056) and somewhat inflated for rows 5–8 (range .059 to .168). These simulation study results, and those in row 4 in particular, provide some justification for the use of missingness model G for the CARDIA data with intermittent missing data patterns. As a complementary approach, we consider models for trends based on maximum likelihood.

5. MAXIMUM LIKELIHOOD ANALYSIS WITH INTERMITTENT MISSING DATA PATTERNS

The most common maximum likelihood (ML) procedure for accommodating missing data in categorical data prob-

lems is the EM algorithm (Dempster, Laird, and Rubin 1977). To study the smoking trends for the four race/sex groups, we used an iteratively reweighted least squares approach originally presented by Galecki et al. (2000) that builds on Molenberghs et al. (1997). The key difference compared to the EM algorithm is that these approaches maximize the observed data likelihood, not a complete data likelihood. Such a mapping can be characterized by applying a linear transformation called *coarsening* to the complete data contingency table to produce the corresponding vector of observed data (Heitjan and Rubin 1991).

Although the ML estimation method of Galecki et al. is similar in spirit to that of Molenberghs et al. in that both approaches maximize the observed likelihood obtained by coarsening the complete data likelihood under a marginal multivariate logistic model, the former improves on the latter in two ways. First, the method of Galecki et al. can accommodate more time points in longitudinal problems. Second, Galecki et al. used an iteratively reweighted least squares technique, which Molenberghs et al. (1997, p. 407) mention may be less tedious computationally than their Newton–Raphson approach. Other likelihood approaches include Baker (1995), who considered the problem where all variables are categorical.

To provide a brief overview of the algorithm of Galecki et al., let π_w be the vector of cell probabilities for the joint distribution of the complete response vector, Y_i , for the w th covariate pattern. In our study we are applying the multivariate logistic model in which π_w is expressed in terms of the function, $\eta(\pi_w)$ (e.g., Dale 1986; Thompson and Baker 1981):

$$\eta(\pi_w) = C_\pi \log(A_\pi \pi_w), \tag{6}$$

where A_π is a matrix of 1s and 0s, C_π is a contrast matrix of -1 s, 0 s, and 1 s. The matrix A_π generates a set of auxiliary probabilities that are contrasted on a logarithmic scale using matrix C_π . We specify a linear model for $\eta(\pi_w)$:

$$\eta(\pi_w) = X_{\pi_w} \beta, \tag{7}$$

where X_{π_w} is a design matrix for a w th covariate pattern. In the analysis of smoking trends, matrix A_π is an 80×16 matrix specified to generate a set of univariate, bivariate, and trivariate marginal probabilities, and joint probabilities in the 2^4 contingency table, that are then contrasted on a logarithmic scale via premultiplication with a 15×80 matrix C_π to obtain a vector of logistic factorial contrasts $\eta(\pi_w)$. These contrasts contain a set of four marginal logits, one for each time point, six marginal log odds ratios for each pair of time points, four trivariate contrasts (e.g., McCullagh and Nelder 1989, p. 220), and one quadruple-variable contrast. Matrices A_π and C_π for the smoking data are the same as matrices A_1 and C_1 , respectively, found in the Appendix of Yang and Becker (1997). Our interest is in the univariate component, or first four elements, of $\eta(\pi_w)$ that corresponds to the logit of the π_i in expression (3); bivariate and higher-order associations are considered nuisance parameters. Compared to WGEE, an attractive feature of ML is that, in the case of MAR, estimates do not depend on the missing data mechanism.

Table 8. Estimated Percent Relative Bias of $\hat{\beta}_1$

Row	(α_3, α_4)	α_2		
		0	-.1	-.2
1	(0, 0)	-.8	-.6	-.8
2	(-1.4, 0)	-.6	-1.9	-3.1
3	(0, .1)	4.0	4.1	3.9
4	(-1.4, .1)	.3	-.8	-2.1
5	(0, .5)	22.9	23.7	24.0
6	(-1.4, .5)	5.0	3.5	1.9
7	(0, 1.0)	45.0	45.1	44.6
8	(-1.4, 1.0)	12.5	11.6	10.3

As in the analyses in sections 3.3 and 4.1 based on GEE and WGEE, our initial models assumed that univariate marginal logits depend on race, sex, and linear and quadratic terms of time. For bivariate odds ratios of marginal probabilities of smoking we considered three structures: independence, all bivariate odds ratios being equal, and unstructured. For all models we assumed that tri- and four-way associations are equal to zero. Likelihood ratio tests show that the best model among those considered was the one that assumes that bivariate odds ratios are unstructured. Furthermore, quadratic effects of time on logits of marginal probabilities were nonsignificant, suggesting that a linear trend for smoking was adequate. Maximum likelihood estimates of slope parameters for each race/sex group are given in Table 7. For methods using an unspecified correlation matrix, the ML approach gives parameter estimates more similar to GEE for blacks and more similar to WGEE for whites. However, qualitatively, all three methods give similar interpretations for the four race/sex groups. Finally, under independence, ML and GEE give the same low estimates, and GEE model-based standard errors (not shown) were the same as those reported for ML.

6. U.S. CENSUS-ADJUSTED SMOKING TREND ESTIMATES

As CARDIA participants have higher attained education, on average, than young adults nationwide (see section 2), the analyses in sections 3, 4, and 5 are not appropriate for drawing inference to the U.S. population of young adults. This section presents an analysis of the CARDIA participants in which, for each race/sex group, models like (1) are fit, but with the subscript *j* indexing four subgroups defined by crossing attained education (less than college degree/college degree) and birth cohort (1963–1967/1955–1962). The four subgroup slope estimates are then combined to produce an overall trend estimate for the U.S. population that is externally adjusted with respect to U.S.

Census data. This is done by taking a weighted average of subgroup specific slope estimates where weights reflect the distribution of the four subgroups in the U.S. population. The particular cut points used to define education and birth cohorts were chosen, in part, because U.S. Census data for these subgroups were readily available. These subgroups, defined as such, capture some of the more salient features in the data as shown in Table 9. For each race/sex group, Table 9 gives the estimated slopes for the four age/education subgroups using each of the four different estimation approaches considered in this study. For WGEE, a missing data model analogous to G in Table 4 was fit but without terms that included race or sex. We applied WGEE with both the monotone estimator (WGEE1) and with the estimator described in section 4, which is applicable to data that includes intermittent missing data patterns (WGEE2). All parameter estimates are based on an unstructured correlation matrix. The four methods give qualitatively similar results; note that the seemingly disparate slope estimates for black males born 1963–1967 and with college degrees have a large variance relative to other subgroups reported in Table 9. There is a significant decrease in smoking among whites females who are college graduates and among the older cohort of white males who are college graduates. All four methods were in general agreement for blacks as no subgroup had a significant decrease in smoking for males or females. Interestingly, GEE gave a significant increase in smoking over the 7-year period for black males with less than a college degree who were in the younger cohort. However, as the MAR-X assumption was violated, we gave the least consideration to the GEE results.

U.S. Census-adjusted slope estimates were determined for each race/sex group as follows. We determined the probability of being in the younger cohort by dividing the proportion of the 1997 U.S. population in the 30–34-year-old group by the proportion ages 30–42 (the latter proportion was determined by multiplying the proportion given

Table 9. Estimates of $\beta_{1j} \times 100$ and Standard Errors by Birth Cohort and Level of Attained Education

Birth cohort	Education	Method	Black males	Black females	White males	White females
1963–1967	No degree	GEE	3.43 (1.47)	.90 (1.13)	1.37 (1.79)	–3.25 (2.15)
		WGEE1	2.30 (1.67)	.99 (1.43)	1.40 (1.91)	–1.53 (2.34)
		WGEE2	2.53 (1.53)	–.07 (1.30)	1.05 (1.83)	–1.92 (2.16)
		ML	3.33 (1.34)	.49 (1.20)	1.28 (1.86)	–2.77 (1.92)
	College graduate	GEE	.76 (4.85)	2.60 (2.80)	–2.55 (4.35)	–6.61 (3.15)
		WGEE1	1.11 (7.14)	2.42 (4.27)	–2.05 (5.02)	–5.95 (3.35)
		WGEE2	–1.95 (7.28)	1.55 (4.45)	–2.05 (4.66)	–5.44 (3.20)
		ML	1.41 (6.48)	3.43 (6.53)	–4.13 (3.54)	–6.45 (3.29)
1955–1962	No degree	GEE	–.16 (.93)	–.92 (.84)	–1.69 (1.09)	–3.39 (1.21)
		WGEE1	–.35 (1.06)	–1.07 (.93)	–1.43 (1.15)	–4.43 (1.30)
		WGEE2	.01 (.96)	–1.46 (.86)	–1.65 (1.12)	–3.58 (1.19)
		ML	–.11 (.98)	–1.06 (1.09)	–1.60 (1.09)	–3.09 (1.15)
	College graduate	GEE	2.02 (2.54)	–1.09 (2.36)	–5.84 (1.65)	–6.64 (1.50)
		WGEE1	2.73 (2.84)	–2.06 (2.75)	–6.34 (1.85)	–7.17 (1.64)
		WGEE2	1.82 (2.81)	.21 (2.56)	–6.18 (1.74)	–7.16 (1.58)
		ML	1.71 (2.44)	–1.41 (2.29)	–5.89 (1.71)	–7.00 (1.64)
U.S. Estimate		GEE	1.25 (.77)	–.15 (.63)	–1.56 (.86)	–4.09 (.91)
		WGEE1	.84 (.90)	–.28 (.75)	–1.46 (.93)	–4.17 (.98)
		WGEE2	.90 (.84)	–.74 (.70)	–1.64 (.89)	–3.80 (.91)
		ML	1.26 (.79)	–.36 (.68)	–1.69 (.84)	–3.85 (.86)

NOTE: All methods used unspecified association structures; empirical standard errors are reported for GEE and WGEE. WGEE1 denotes the monotone estimator, and WGEE2 the intermittent data estimator. The U.S. estimate is the externally adjusted (weighted) estimate of smoking trend for young adults in the United States.

for 35–44-year-olds by .8 and adding the result to the proportion for the 30–34-year-old group (U.S. Bureau of the Census, 1998b). Next, we obtained the probability of having at least a college degree given birth cohort (from Census data (U.S. Bureau of the Census, 1998a) we abstracted from column “25–29-year-olds” the 1992 entry for the younger cohort and the average of the 1984, 1985, 1986, and 1987 entries for the older cohort). Finally, we applied Bayes’s theorem to obtain the joint probability distribution of education and birth cohort among young adults expressed as a vector $\mu' = (\mu_1, \mu_2, \mu_3, \mu_4)$ and satisfying $\sum_{j=1}^4 \mu_j = 1$. U.S. Census-adjusted slope estimates were computed as $\hat{\beta}_{\text{adj}} = \mu' \hat{\beta}_1$, where $\hat{\beta}_1 = (\hat{\beta}_{11}, \hat{\beta}_{12}, \hat{\beta}_{13}, \hat{\beta}_{14})$, corresponding to the subgroups in the order in which they appear in Table 9. The standard error of $\hat{\beta}_{\text{adj}}$ is computed as the square root of $\mu' \text{cov}(\hat{\beta}_{\text{adj}}) \mu$. For all four methods in Table 9, we set μ to (.34, .05, .54, .07), (.35, .04, .54, .07), (.27, .09, .49, .15), and (.27, .09, .50, .14) for black males, black females, white males, and white females, respectively. The U.S. Census-adjusted slopes are generally larger than the unadjusted slopes in Table 7 (Table 6 for the monotone WGEE(G)-Unstructured estimator). Unadjusted slopes tend to be inappropriately small because the CARDIA cohort contained a disproportionate representation of well-educated individuals. Using an unstructured correlation model, adjusted estimates for WGEE and ML gave similar conclusions, namely that the only significant trend in cigarette smoking from 1986 to 1993 was a decline in smoking among white females. Whereas the unadjusted slope for white males was significantly less than zero, the U.S. Census-adjusted slope was nonsignificant.

7. DISCUSSION

This article examines the use of GEE, WGEE, and ML techniques in the estimation of trends in cigarette smoking in a cohort of young adults. The primary approach was to use a monotone WGEE estimator because the WGEE and ML intermittent missing data approaches require stronger assumptions about the missing data mechanism. Additionally, in the CARDIA study, smokers were more likely to drop out than nonsmokers and so a GEE analysis that ignored the missing data mechanism provided biased slope estimates that were generally too small. This statement was particularly true when the independence working correlation was used in GEE. Application of the WGEE monotone estimator for MAR data (Robins et al. 1995) revealed that over a 7-year period the prevalence in cigarette smoking among a cohort of young adults, standardized to U.S. distributions for age and education, did not change significantly for black males, black females, and white males. Only for white females did smoking significantly decrease. Application of a simple WGEE intermittent data estimator and an ML estimator supported these conclusions. An earlier WGEE analysis of CARDIA smoking trends based on all five time points (Wagenknecht et al. 1998) reported a significant increase in smoking among black males, and although the ordering of the magnitudes of the trends among the four race/sex groups was the same as found here, the

reported slopes were larger than ours; they were based mistakenly on biased estimating functions that had the effect of giving inappropriately large slope estimates.

This article presented methods for estimation of parameters in marginal regression models for incomplete longitudinal binary data that are valid when data are missing at random. There has also been much work on the nonignorable missingness problem including maximum likelihood (Baker 1995; Molenberghs and Goetghebeur 1997) and estimating equation (Rotnitzky, Robins, and Scharfstein 1998) approaches where missingness may depend directly on unobserved outcomes, i.e., selection models. Several authors (Follmann and Wu 1995; Ten Have, Kunselman, Pulkstenis, and Landis 1998) have proposed random effects models for longitudinal data that account for nonignorable missing data indirectly by specifying models for primary response and missingness that share a common random parameter. Because these shared parameter models depend on the missing outcomes implicitly, their specification for dropouts applies, unmodified, to intermittent missing data. The common problem with all approaches for nonignorable missing data is underidentifiability of parameters, and that inference is dependent on model assumptions that cannot be checked as they involve unobserved data (Little 1995). With respect to either MAR or NI approaches, sensitivity analyses are suggested to assess the effect of alternative assumptions about the missing data on inferences on parameters in the measurement model.

The data analyst has various considerations when choosing a statistical method for incomplete binary longitudinal data when MAR-Y missingness patterns are suspected. The WGEE procedure for dropouts considered here extends more easily to large cluster sizes than the ML approach, although the method of Galecki et al. (2000) provides more flexibility than its predecessors. Unlike ML approaches under MAR-Y, WGEE regression parameter estimates in the trends model are affected by inclusion of covariates in the dropout model that are not in the trends model. In contrast, in ML there is no need to model the missing data under MAR-Y, because estimation of the MAR-Y dropout model does not affect estimation of the trends model (Little and Rubin 1987). To obtain ML estimates, the NLMEM SAS/IML macro was used (Galecki 1998). Although NLMEM is designed to fit hierarchical models, it can be used in the context of incomplete categorical data as was considered in this article. WGEE parameter estimates may be obtained using commercially available software that allows for observations to be weighted. For example, after obtaining appropriate weights for the missing data, one could apply SAS PROC GENMOD (SAS Institute 1997) with the SCWGT option. Obtaining standard errors (4) requires additional programming.

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